



VISAYAS  
STATE UNIVERSITY



## Lesson 2.3 - Inequalities

Department of Mathematics  
Visayas State University



At the end of the lesson, the students will be able to:

- Define linear and nonlinear inequalities.
- Solve linear inequality.
- Solve nonlinear inequality.
- Define absolute value inequality.
- Solve absolute value inequality

# Inequality

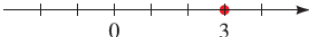
Some problems in algebra lead to inequalities instead of equations. An inequality looks just like an equation, except that in the place of the equal sign is one of the symbols,  $<$ ,  $>$ ,  $\leq$ , or  $\geq$ .

## Definition

An **inequality** is any sentence having one of the words  $<$ ,  $>$ ,  $\leq$ , or  $\geq$ . A **solution** to an inequality is any number making it a true sentence. The set of all solutions is called the **solutions set**. When we have found the set of all solutions to an inequality, we say that we have **solved the inequality**.

# Inequality

To solve an inequality that contains a variable means to find all values of the variable that make the inequality true. Unlike an equation, an inequality generally has infinitely many solutions, which form an interval or a union of intervals on the real line.

|                              | <b>Solution</b> | <b>Graph</b>                                                                       |
|------------------------------|-----------------|------------------------------------------------------------------------------------|
| Equation: $4x + 7 = 19$      | $x = 3$         |  |
| Inequality: $4x + 7 \leq 19$ | $x \leq 3$      |  |

## RULES FOR INEQUALITIES

### Rule

1.  $A \leq B \Leftrightarrow A + C \leq B + C$
2.  $A \leq B \Leftrightarrow A - C \leq B - C$
3. If  $C > 0$ , then  $A \leq B \Leftrightarrow CA \leq CB$
4. If  $C < 0$ , then  $A \leq B \Leftrightarrow CA \geq CB$
5. If  $A > 0$  and  $B > 0$ ,  
then  $A \leq B \Leftrightarrow \frac{1}{A} \geq \frac{1}{B}$
6. If  $A \leq B$  and  $C \leq D$ ,  
then  $A + C \leq B + D$
7. If  $A \leq B$  and  $B \leq C$ , then  $A \leq C$

### Description

**Adding** the same quantity to each side of an inequality gives an equivalent inequality.

**Subtracting** the same quantity from each side of an inequality gives an equivalent inequality.

**Multiplying** each side of an inequality by the same *positive* quantity gives an equivalent inequality.

**Multiplying** each side of an inequality by the same *negative* quantity *reverses the direction* of the inequality.

**Taking reciprocals** of each side of an inequality involving *positive* quantities *reverses the direction* of the inequality.

Inequalities can be added.

Inequality is transitive.

# Linear Inequality

## Definition

An inequality is **linear** if each term is constant or a multiple of the variable.

To solve a linear inequality, we isolate the variable on one side of the inequality sign.

## Example

Solve each inequality and sketch the solution set.

❶  $3x < 9x + 4$

❷  $16 - 7y \geq 10y - 4$

❸  $0.4x + 5 \leq 1.2x - 4$

❹  $4 \leq 3x - 2 \leq 13$

❺  $-2 < x + 2 < 8$

❻  $6 > -x \geq -2$

❼  $-18 \leq -2x - 7 < 0$

# Nonlinear Inequality

To solve inequalities involving squares and other powers of the variable, we use factoring, together with the following principle.

## Theorem (The Sign of a Product or Quotient )

- *If a product or a quotient has an even number of negative factors, then its value is positive.*
- *If a product or a quotient has an odd number of negative factors, then its value is negative.*

## Example

Solve the following inequality.

①  $x^2 \leq 5x - 6$

②  $x(x - 1)^2(x - 3) < 0$

③  $\frac{1 + x}{1 - x} \geq 1.$

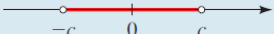
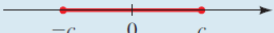
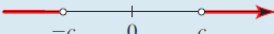
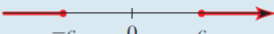
## GUIDELINES FOR SOLVING NONLINEAR INEQUALITIES

1. **Move All Terms to One Side.** If necessary, rewrite the inequality so that all nonzero terms appear on one side of the inequality sign. If the nonzero side of the inequality involves quotients, bring them to a common denominator.
2. **Factor.** Factor the nonzero side of the inequality.
3. **Find the Intervals.** Determine the values for which each factor is zero. These numbers will divide the real line into intervals. List the intervals that are determined by these numbers.
4. **Make a Table or Diagram.** Use **test values** to make a table or diagram of the signs of each factor on each interval. In the last row of the table determine the sign of the product (or quotient) of these factors.
5. **Solve.** Use the sign table to find the intervals on which the inequality is satisfied. Check whether the **endpoints** of these intervals satisfy the inequality. (This may happen if the inequality involves  $\leq$  or  $\geq$ .)

# Absolute Value Inequality

We use the following properties to solve inequalities that involve absolute value.

## PROPERTIES OF ABSOLUTE VALUE INEQUALITIES

| Inequality      | Equivalent form           | Graph                                                                              |
|-----------------|---------------------------|------------------------------------------------------------------------------------|
| 1. $ x  < c$    | $-c < x < c$              |  |
| 2. $ x  \leq c$ | $-c \leq x \leq c$        |  |
| 3. $ x  > c$    | $x < -c$ or $c < x$       |  |
| 4. $ x  \geq c$ | $x \leq -c$ or $c \leq x$ |  |

## Example

Solve the inequality

①  $|x - 5| < 2$

②  $|3x + 2| \geq 4$